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INTERDISCIPLINARY PROJECT PHASE 2 PRESENTATION

<u>TITLE:</u> FoodSense: Multimodal AI System for Early Spoilage Prediction

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INTRODUCTION

- Food spoilage is a major challenge in households, restaurants, and food storage systems, leading to health risks and significant food waste. Improper storage conditions such as high temperature, humidity, and bacterial activity accelerate food degradation.
- Traditional food containers provide no real-time information about food freshness, forcing users to rely on manual inspection, which often fails to detect early spoilage.
- This project proposes FoodSense, a multimodal AI-based food monitoring system that combines sensors and computer vision to detect early signs of spoilage and alert users in real time.
- Supports SDG 12 (Responsible Consumption) by reducing food waste and SDG 3 (Good Health) by improving food safety.

LITERATURE SURVEY

Journal, Year	Paper Title	Methodology Adopted	Research Gaps
ScienceDirect (2025)	From lab to market: Real-time food safety monitoring via spectroscopy, blockchain and artificial intelligence	Focus: Spectroscopy + AI + IoT + blockchain integration Contribution: Replaces slow lab testing with real-time monitoring systems Key idea: Farm-to-fork continuous quality tracking	Lack of **large-scale real-world validation** No standard framework integrating all 3 technologies seamlessly Most studies remain prototype-level
Journal of Food and Biotechnology (2025)	Next-Generation Food Safety Monitoring Using Biosensors and Internet-of-Things Technologies	Focus: Biosensors + IoT platforms Applications: Smart packaging, pathogen detection, cold-chain monitoring Key idea: Shift from reactive → proactive real-time food safety	Limited **sensor accuracy in dynamic environments** High cost → not scalable for small farmers Adoption barrier in developing regions
International Journal for Research in Applied Sciences and Engineering Technology (2023)	Blockchain and IoT based Food Traceability for Smart Agriculture	Focus: Integrated IoT sensors + blockchain Contribution: Automated tracking across production, transport, storage Key idea: Reduces human intervention, improves transparency	Weak **interoperability between IoT devices and blockchain** Lack of standard communication protocols

LITERATURE SURVEY

Journal, Year	Paper Title	Methodology Adopted	Research Gaps
ScienceDirect (2025)	Recent trends in blockchain traceability of food products	Focus: Blockchain + AI + IoT integration Key idea: Ensures data integrity and transparency across supply chains Relevance: Strong foundation for bio-traceability systems	Insufficient research on **data governance & ownership** Lack of regulatory frameworks
MDPI (2025)	Advanced Digital Solutions for Food Traceability: NIRS, RFID, Blockchain, IoT	Focus: Multi-technology integration Key idea: Real-time monitoring + automation reduces errors and improves trust	Fragmented systems → no **unified architecture** Data silos across technologies
PubMed (2023)	Redefining food safety traceability system through blockchain	Focus: Blockchain-enabled Food Safety Traceability Systems (FSTS) Key idea: Full lifecycle tracking from producer to consume	Limited **performance analysis under heavy data loads** Energy consumption concerns

LITERATURE SURVEY

Journal, Year	Paper Title	Methodology Adopted	Research Gaps
Royal Society of Chemistry (2025)	Comprehensive review on the integration of self-healing polymers with smart food traceability: scope, application and challenges	Focus: Smart packaging + RFID + blockchain Key idea: Durable traceability systems that survive supply chain damage	Early-stage research → lacks commercial feasibility studies Durability and cost issues
ArXiv (2025)	Smartphone-Based Food Traceability System Using NoSQL Database	Focus: RFID + mobile sensors Features: Real-time monitoring of temperature, humidity, location Key idea: Affordable real-time bio-traceability solution (case: kimchi supply chain)	Limited security & privacy protection mechanisms Dependency on user participation (manual input errors)
ArXiv (2017)	Improving risk management by using smart containers for real-time traceability	Focus: Smart containers + IoT Key idea: Real-time logistics data improves food quality control decisions	Lack of predictive analytics (AI integration) Focus is monitoring, not decision-making

LITERATURE SURVEY

Journal, Year	Paper Title	Methodology Adopted	Research Gaps
IEEE Xplore (2021)	Blockchain in Food Traceability: A Systematic Literature Review	Focus: Review of blockchain-based traceability systems Key idea: Transparency + data integrity in food supply chains	Most studies are **theoretical** Lack of empirical datasets and benchmarking
IEEE Xplore (2019)	Blockchain-Driven IoT for Food Traceability With an Integrated Consensus Mechanism	Focus: IoT sensors + blockchain consensus Key idea: Real-time monitoring with tamper-proof records	Consensus mechanisms cause **latency in real-time systems** Trade-off between speed vs security
IEEE Xplore (2022)	Food Supply Chain Traceability System Using Blockchain Technology	Focus: Practical implementation Key idea: Farm-to-consumer tracking using distributed ledger	Weak **integration with legacy systems (ERP, SCM tools)** Adoption barriers in existing industries

PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Food spoilage remains a major issue in households and food storage systems due to the absence of real-time monitoring and early detection mechanisms, leading to health risks and increased food waste. Traditional storage methods fail to track critical factors such as temperature, humidity, gas emissions, and visual changes that indicate deterioration. Therefore, there is a need for an intelligent, cost-effective system that integrates sensor data and computer vision to continuously monitor food conditions, predict early spoilage, and provide timely alerts to improve food safety and reduce waste.

Primary Objectives

- Design and develop a multimodal system integrating gas, temperature, humidity sensors, and computer vision for food spoilage detection
- Enable continuous real-time monitoring of food storage conditions with timely alert mechanisms
- Develop and train AI/ML models to accurately predict early stages of food spoilage

Secondary Objectives

- Apply explainable AI techniques such as Grad-CAM, LIME, and saliency maps to interpret model decisions
- Optimize the system for low cost, energy efficiency, and practical deployment
- Evaluate and validate system performance using experimental data for accuracy and reliability

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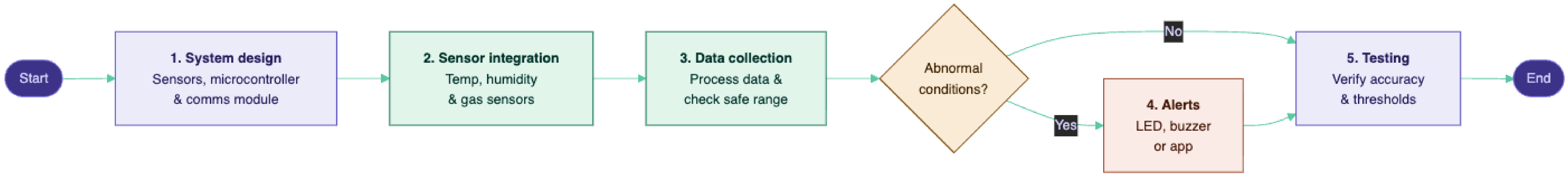
METHODOLOGY



1. Identification of Spoilage Indicators

- Monitor key parameters:
    - Temperature
    - Humidity
    - Gas emissions (VOCs using MQ sensors)
    - Visual changes (color, texture, mold growth)

2. System Architecture Design

- Design an integrated system including:
    - Environmental sensors
    - Camera module
    - Microcontroller (ESP32/Arduino)
    - AI processing unit (local/cloud)
- Define data flow between hardware and AI modules

3. Camera Module Integration

- Attach:
    - ESP32-CAM or USB camera
- Ensure:
    - Fixed positioning
    - Consistent lighting conditions
- Capture images at regular intervals
- Store or transmit images for processing
- ○

METHODOLOGY

4. Data Acquisition and Transmission

- Program ESP32 using Arduino IDE
- Perform:
  - Periodic sensor data collection
  - Analog-to-digital conversion
  - Timestamping of data
- Transmit data using:
  - WiFi (HTTP/MQTT protocols)
- Store data in:
  - CSV files or databases

5. Computer Vision Model Development

- Use PyTorch framework
- Select model: CNN or Vision Transformer
- Fine-tune pretrained models
- Train classifier for spoilage stages
- Validate using training/testing split

6. Sensor Data Processing

- Treat sensor data as time-series
- Extract features:
  - Gas concentration trends
  - Temperature and humidity variations
  - Rate of change in readings
- Normalize features before model input

7. Multimodal Data Fusion

- Combine:
  - Image features and Sensor features
- Use neural network layers for fusion
- Train unified model for improved prediction
- Output: Spoilage prediction score

8. AI Integration

- Apply:
  - Grad-CAM for image interpretation
  - LIME/SHAP for sensor feature importance
- Identify key factors influencing predictions

METHODOLOGY

10. Alert and Decision System

- Define spoilage thresholds
- Trigger alerts when:
  - Unsafe conditions detected
- Notification methods:
  - LEDs / Buzzers
  - Mobile/Web interface
- Display real-time freshness status

11. System Integration

- Integrate:
  - Hardware sensing module
  - AI prediction pipeline
- Ensure:
  - Continuous data flow
  - Synchronization between sensing, processing, and alerts

12. **Prototype Development and Testing**

Incorporation of Phase 1 Suggestions

Tools & Techniques

Hardware :

- **Arduino uno microcontroller** - Used for data from real time environment
- **MQ135 Gas sensor** - Detect gases such as ammonia (NH3) and carbon dioxide (CO2) in the environment
- **DHT22 Humidity sensor** - Check the temperature and humidity of the environment in which the food is present
- Jumper wires and bread board

Software :

- **Arduino IDE** - for powering and sending instructions to the microcontroller
- **ML Model**

Work Progress so far

Implemented model on practical scale
Collected data from food rotting

Partial Results

Work to be done for Completion

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Thank You!